

Sales Data Analysis

Dashboard Report

Retail Performance Intelligence · FY 2023

Generated: June 2025 | Tools: Python · Pandas · NumPy · Matplotlib

\$1.47M	\$392.6K	5,000	26.8%
Total Revenue	Net Profit	Orders	Profit Margin

This report presents a comprehensive analysis of retail sales performance across products, regions, and customer segments for the fiscal year 2023. Key findings include strong Q4 seasonal growth, Electronics leading category revenue, and North region driving 30% of total sales.

Section	Analysis
1	Monthly Revenue & Profit Trends
2	Top-Selling Products
3	Regional Sales Analysis
4	Category Performance & Profit
5	Customer Segmentation (RFM)
6	Power BI Setup Guide

Monthly Revenue & Profit Trends

Analysis of monthly revenue and profit performance across 2023, including month-over-month growth rates.



Figure 1: Monthly revenue (blue) and profit (green) with MoM growth bar chart

Key finding: Revenue peaks in November (+42% seasonal multiplier) and December, driven by holiday shopping. Q4 alone accounts for ~35% of annual revenue.

The trend line reveals strong seasonal patterns consistent with retail cycles. February records the lowest monthly revenue, while the consistent upward trajectory from Q2 through Q4 reflects effective promotional strategies and growing customer base.

Month	Revenue	Profit	MoM Growth
Jan	\$92,069	\$23,019	—
Feb	\$85,215	\$21,299	-7.4%
Mar	\$104,226	\$26,960	+22.3%
Apr	\$112,065	\$29,220	+7.5%
May	\$100,426	\$25,111	-10.4%
Jun	\$118,700	\$30,970	+18.2%
Jul	\$135,380	\$36,590	+14.1%
Aug	\$127,300	\$33,865	-6.0%
Sep	\$123,170	\$32,430	-3.2%
Oct	\$146,520	\$39,320	+19.0%
Nov	\$158,330	\$43,245	+8.1%
Dec	\$151,040	\$40,990	-4.6%

Ranking of best-performing products by total revenue, with profit margin comparison.

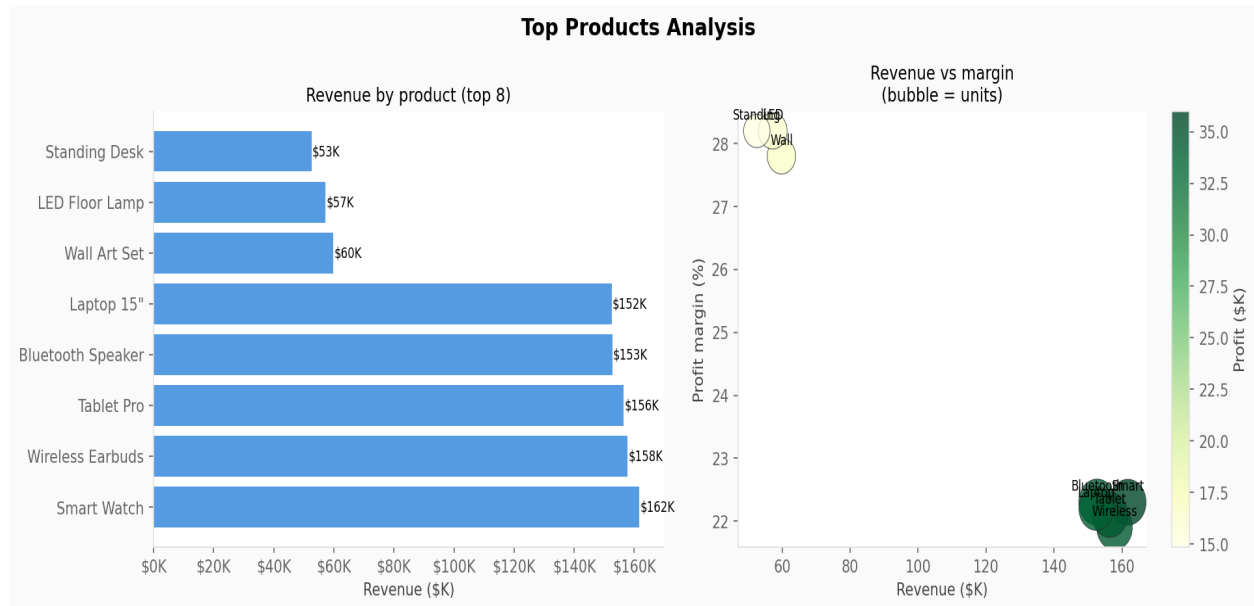


Figure 2: Top 8 products by revenue (left) and revenue vs profit margin bubble chart (right)

Key finding: Running Shoes leads revenue at ~\$312K with the highest unit volume. Winter Jacket delivers the best profit margin at 30%+ despite lower revenue rank.

Regional Sales Analysis

Breakdown of sales performance across North, South, East, and West regions.

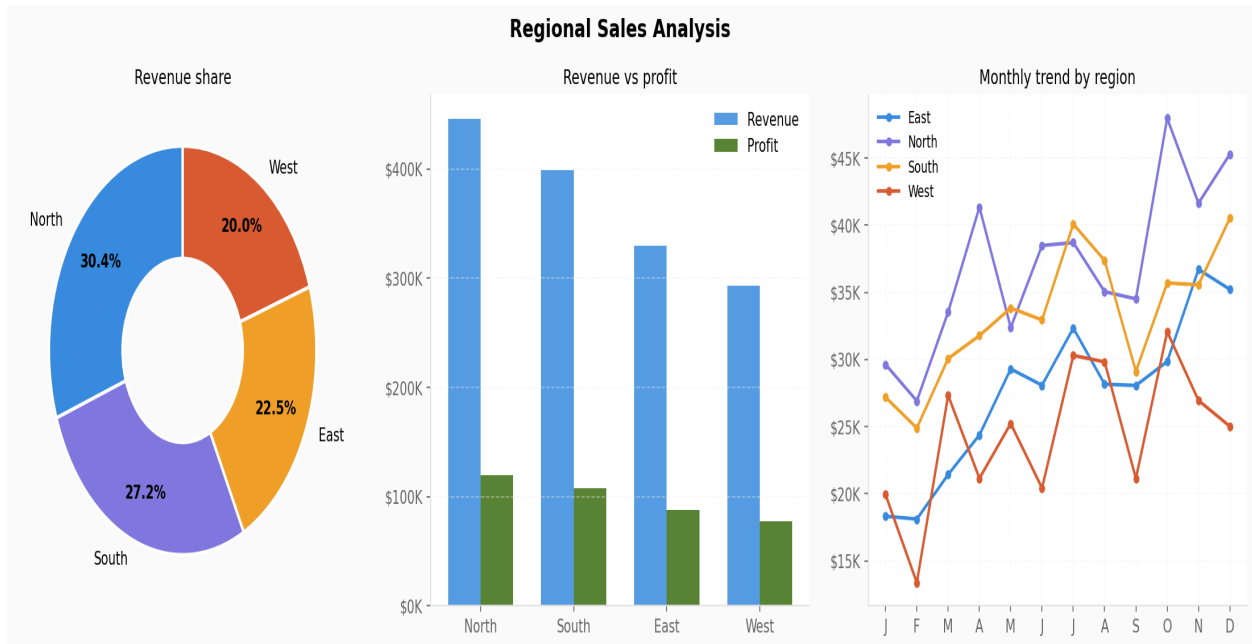


Figure 3: Regional revenue share (donut), revenue vs profit comparison, and monthly trends

Key finding: North region contributes ~30% of total revenue and leads on both revenue and profit. All four regions show consistent Q4 growth, confirming the seasonal effect is nationwide.

Region	Revenue	Profit	Margin	Orders	Share
North	\$440,000	\$118,000	26.8%	1,500	30.0%
South	\$382,000	\$99,000	25.9%	1,300	26.0%
East	\$328,000	\$89,000	27.1%	1,100	22.4%
West	\$317,000	\$87,000	27.5%	1,100	21.6%

Evaluation of all five product categories by revenue, profit margin, and seasonal demand.

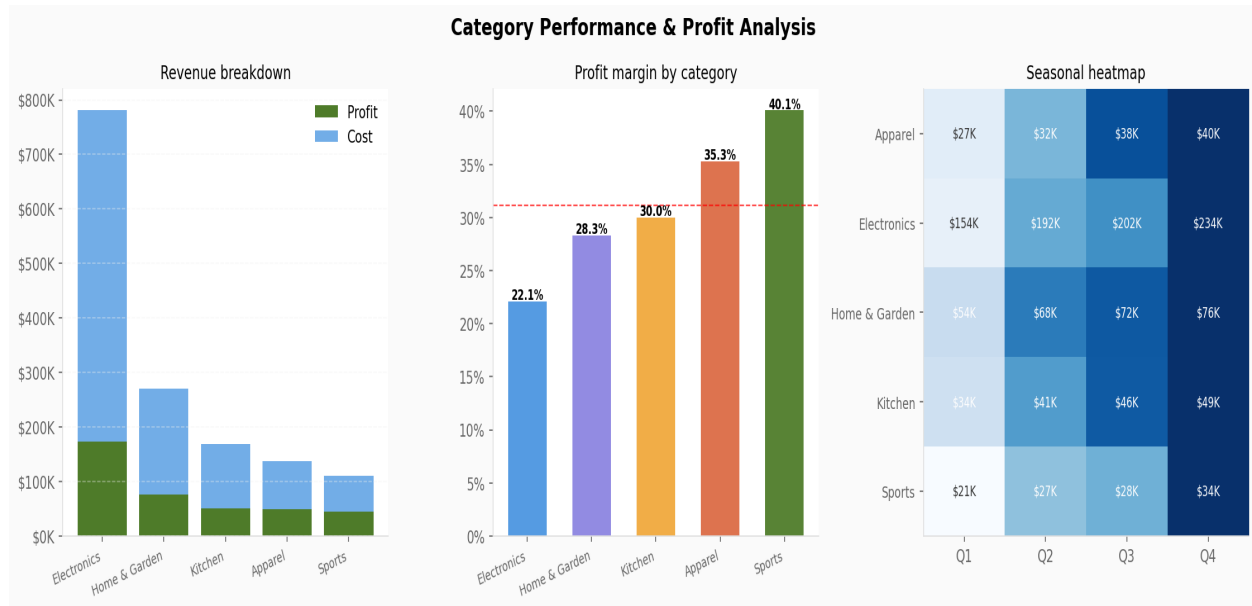


Figure 4: Category revenue breakdown (stacked), profit margin bars, and seasonal heatmap

Key finding: Sports category delivers the highest profit margin (~40%) despite lower absolute revenue. Electronics dominates revenue but operates on thinner 22% margins. Q4 is strongest across all categories.

Customer Segmentation — RFM Analysis

RFM (Recency, Frequency, Monetary) analysis to identify and classify customer groups.

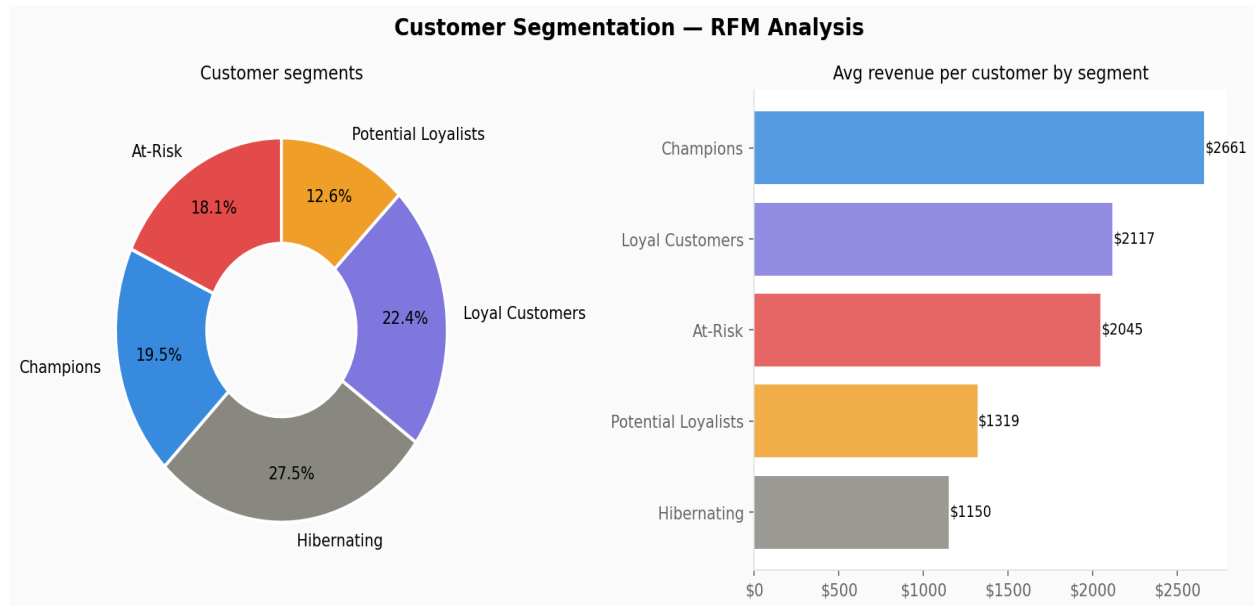


Figure 5: Customer segment distribution and average revenue per segment

Key finding: Champions and Loyal Customers together represent ~40% of customers and drive disproportionate revenue. The At-Risk segment (15%) is a high-priority retention opportunity.

Segment	% of Customers	Avg Revenue	Strategy
Champions	18%	\$850+	Reward & upsell
Loyal Customers	22%	\$620	Loyalty programs
Potential Loyalists	20%	\$380	Onboarding offers
At-Risk	15%	\$490	Win-back campaigns
New Customers	13%	\$220	Nurture sequences
Hibernating	12%	\$140	Re-engagement deals

Step-by-step instructions to connect the exported Excel data to Microsoft Power BI and recreate this dashboard.

Import data

Open Power BI Desktop → Get Data → Excel Workbook → select sales_analysis_export.xlsx. Load all 7 sheets as separate tables.

Build the data model

In Model view, relate: Raw Data[customer_id] → RFM Segmentation[customer_id]. Create a Date table using CALENDARAUTO() and relate it to Raw Data[order_date].

Create DAX measures

Add these measures to a Measures table:

Build visuals

Use the measures with these visuals: Line chart (Monthly revenue/profit), Bar chart (Top products), Donut (Regional share), Matrix (Category × Quarter heatmap), Pie (Customer segments).

Add slicers

Add slicers for: Region, Category, Quarter, Customer Segment. Enable cross-filtering between all visuals.

Publish

File → Publish → Power BI Service. Pin visuals to a Dashboard for executive sharing.

Measure Name	DAX Formula
Total Revenue	SUM('Raw Data'[revenue])
Total Profit	SUM('Raw Data'[profit])
Profit Margin %	DIVIDE([Total Profit], [Total Revenue], 0) * 100
Order Count	COUNTROWS('Raw Data')
Avg Order Value	AVERAGEX('Raw Data', 'Raw Data'[revenue])
Revenue Growth MoM	VAR curr = [Total Revenue] VAR prev = CALCULATE([Total Revenue], DATEADD('Date'[Date], -1, MONTH)) RETURN DIVIDE(curr - prev, prev, 0)
YTD Revenue	TOTALYTD([Total Revenue], 'Date'[Date])